

# Why Option Prices Can Be Written as Fourier Integrals

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### 1. Starting From the Risk-Neutral Expectation

For a European call option maturing at  $T$  with strike  $K$ , the price is

$$C(K) = e^{-rT} E[(S_T - K)^+].$$

Let us write the log-price as

$$X_T = \log S_T.$$

Then

$$S_T = e^{X_T}, \quad (S_T - K)^+ = (e^{X_T} - K)^+.$$

Expanding the expectation using the risk-neutral density  $f_{X_T}(x)$ :

$$C(K) = e^{-rT} \int_{\log K}^{\infty} (e^x - K) f_{X_T}(x) dx.$$

This shows that if we know the density of  $X_T$ , then pricing is simply an integral. The *Fourier approach* replaces the density by its Fourier-transform representation.

### 2. Why the Density Can Be Written as an Inverse Fourier Integral

Let  $\phi_{X_T}(u)$  be the characteristic function of  $X_T$ :

$$\phi_{X_T}(u) = E[e^{iuX_T}].$$

The Fourier inversion theorem states that if  $X_T$  has a density  $f_{X_T}(x)$ , then

$$f_{X_T}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_{X_T}(u) e^{-iux} du.$$

This is a fundamental harmonic-analysis result: the characteristic function uniquely determines the probability density, and the density can be recovered by taking the inverse Fourier transform of the characteristic function.

### 3. Substituting the Fourier Representation of the Density

Insert the inverse Fourier integral for the density into the option pricing formula. Start from

$$C(K) = e^{-rT} \int_{\log K}^{\infty} (e^x - K) f_{X_T}(x) dx.$$

Replace  $f_{X_T}(x)$ :

$$C(K) = e^{-rT} \int_{\log K}^{\infty} (e^x - K) \left[ \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_{X_T}(u) e^{-iux} du \right] dx.$$

Interchange integrals (justified under mild integrability conditions):

$$C(K) = \frac{e^{-rT}}{2\pi} \int_{-\infty}^{\infty} \phi_{X_T}(u) \left[ \int_{\log K}^{\infty} (e^x - K) e^{-iux} dx \right] du.$$

Now the option price appears as a *Fourier transform* of a known kernel times the characteristic function.

This is the key step:

**Option prices can be expressed as Fourier integrals because the density itself is the inverse Fourier transform of the characteristic function.**

### 4. Cleaning the Expression

Define  $k = \log K$ . The inner integral has closed form for all  $u$ . Thus the entire pricing formula becomes

$$C(K) = \frac{e^{-rT}}{2\pi} \int_{-\infty}^{\infty} e^{-iuk} \psi(u) du,$$

where  $\psi(u)$  is an explicit expression built from  $\phi_{X_T}$ . This is exactly a Fourier transform.

### 5. Why FFT Is Useful

The expression above has the generic structure

$$\text{Price}(K) = \int e^{-iuk} G(u) du,$$

which is a Fourier transform of the function  $G(u)$ . Once written in this form:

- the integral becomes a discrete Fourier transform (DFT),
- which can be computed by the fast Fourier transform (FFT),

- allowing hundreds of strikes to be priced in one shot.

Thus the FFT is not the model itself: it is simply the computational engine used because the pricing formula reduces to a Fourier transform.